1. Clinical Question & Core Hypothesis

Clinical Question

Tier 1 analysis identifies a tension between apparent local immune organization (PD-L1/TLS signals) and poor clinical outcomes under standard therapy — why do immune-structured tumors sometimes fail to derive benefit from cytotoxic therapy in localized high■risk prostate cancer?

Working Hypothesis

Primary mechanistic hypothesis: In localized high-risk prostate cancer, TLS-like lymphoid aggregates indicate organized anti-tumor immunity, but their effectiveness is contingent on tumor-intrinsic PGE2/EP4 signaling. When PTGER4 expression is high within tumor and/or immune cells, PGE2-EP4 signaling upregulates PD-L1 (CD274), promoting T-cell exhaustion and damping TLS-driven cytotoxic responses, thereby reducing docetaxel sensitivity. Clinically, TLS-high tumors with concurrent high PTGER4/CD274 expression will exhibit diminished therapy_response to docetaxel relative to TLS-low or PTGER4/CD274-low tumors. Secondary hypothesis: pharmacologic co-targeting of EP4 (PTGER4) and PD-L1 can restore TLS-mediated anti-tumor activity and enhance docetaxel response in TLS-high, PTGER4/CD274-high tumors. In addition, genomic risk (Decipher_GC), cell-cycle signals (CENPA), and inflammatory/urinary biomarkers, as secondary modifiers, will modulate this axis by affecting baseline tumor aggressiveness and the inflammatory milieu, thereby improving the predictive model without being primary causal drivers. Endpoint specification: therapy_response will be defined in the neoadjuvant/localized setting by objective measures of docetaxel sensitivity (e.g., radiographic/biochemical response, pathologic tumor viability) and will be correlated with TLS status, PTGER4/CD274 expression, and modifier biomarkers.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	PTGER4, CD274
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

PTGER4: PTGER4 → PD■L1 → T■cell exhaustion path (Core Path 1) because it provides a mechanistic immunosuppressive axis; on the Decipher GC → survival/therapy path (Core Path 2) as a genomic-risk context that can modulate outcomes; and on the PD■L1 prognostic link to docetaxel response (Core Path 3) to tie immune state to chemotherapy effectiveness.

CD274: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	Decipher_GC	Supporting modifier from knowledge graph

Secondary	CENPA	Supporting modifier from knowledge graph
Secondary	inflammation_biomarkers	Supporting modifier from knowledge graph
Secondary	urinary_biomarkers	Supporting modifier from knowledge graph
Secondary	radiomic_features	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: mediates PGE2-driven immunosuppression and has been linked to increased PD-L1 expression in various cancers; high PTGER4/CD274 activity within an otherwise TLS-rich microenvironment provides a plausible mechanistic bridge explaining observed discordance between TLS presence and docetaxel response in localized prostate cancer.

Prediction 2: At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

Prediction 3: In functional experiments, restoring or enhancing the anchor gene's signaling will improve immune-mediated tumor control in imaging-feature-positive models.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Multi-modal validation pipeline: computational -> spatial -> functional, providing mechanistic depth beyond association.

Caveats & Risks

- Gene expression does not equal protein activity; phospho-protein or functional assays are needed for confirmation.

Alternative Directions

- Decipher_GC: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- CENPA: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.